Field Hours

CSCE 630 Speech Processing

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

The source and the vocal-tract filter

You model a vowel as a source passed through a filter. The source for voiced sound is a pulse train, often near 120 hertz for one speaker and 200 for another. The vocal tract is the filter: it boosts some harmonics and cuts others, and those peaks are the formants. Change the filter and you change the vowel. Change the pulse rate from 120 to 150 and you change the pitch, not the vowel.

An unvoiced sound has no pulse train. You replace the source with noise and keep a filter, which is how a fricative differs from a vowel at the same mouth shape. A whisper is the same move: drop the pulses, keep the formants. In a toy synth you can emit 120 impulses in one second, or noise, and pass either through the same 2 formant peaks. If the peaks stay put, the vowel identity stays put.

Pitch and vowel are different knobs.

Source

Pulses near 120 hertz

Or noise, for unvoiced

Sets the pitch

Filter

Formant peaks

Same peaks, same vowel

Mouth shape, not rate

Harmonics of a 100 hertz source

1. A pulse train at 100 hertz has energy at 100, 200, 300, and 400 hertz, the integer multiples.

2. A filter peak near 300 hertz boosts the third harmonic and leaves the 100 hertz line weaker. That pattern is the vowel.

3. Move the pulse rate to 125 hertz and the lines sit at 125, 250, 375, and 500. If the peak stays near 300, the vowel is the one you kept and the pitch is the one you changed.

Problem 1. You raise the pulse rate from 120 hertz to 150 and leave the formant peaks fixed. What changed?

Answer. The pitch. The vowel is the filter. A new pulse rate moves the harmonics and does not by itself move the formants.

Problem 2. A whisper keeps the same 2 formant peaks but replaces 120 impulses per second with noise. Which sound class did you leave, and which did you keep?

Answer. You left voiced speech, because the pulse train is gone. You kept the vowel identity, because the filter peaks stayed put.

Lecture 2 · 90 minutes

What the ear groups together

You do not hear a spectrum as a flat list of bins. The ear groups frequencies into critical bands that widen as you go up: near 500 hertz a band is about 100 hertz wide, and near 4000 hertz it is wider. Masking follows those bands. A loud tone can hide a quieter neighbor inside the same band and leave a tone two bands away easy to hear. A feature built on linear bins fights that grouping.

The mel scale is a practical map of that widening. You space filter centers linearly below about 1000 hertz and logarithmically above it, then put a triangle filter on each center. With 26 filters you cover a speech band from 0 to 8000 hertz at a 16 kilohertz sample rate. Energy in one triangle is one number. A loud bin outside the triangle contributes 0, which is the model of masking you are building, not a picture of the cochlea.

From linear bins to one band energy.

Start from hertz bins -> Place 26 triangles -> Weight bins inside each -> Sum energy per triangle

Two bins, one triangle

1. A triangle peaks at 1000 hertz and hits 0 at 800 and at 1200. A bin at 1000 has weight 1. A bin at 1400 has weight 0.

2. Bin energies are 4 at 1000 hertz and 10 at 1400. The triangle output is 1 times 4 plus 0 times 10, so 4.

3. The loud bin did not mask its way into this channel, because it sits outside the triangle. A second triangle centered near 1400 would be the place that 10 belongs.

Problem 1. Triangle weights are 1 and 0 on bin energies 4 and 10. What energy does this channel report?

Answer. 4. The bin of energy 10 has weight 0, so it adds nothing.

Problem 2. Why space the 26 filters linearly below about 1000 hertz and more widely above it?

Answer. Critical bands are narrower near 500 hertz, about 100 hertz, and wider near 4000. The mel spacing is a practical map of that widening, not a uniform grid of bins.

Lecture 3 · 90 minutes

Frames, hops, and a window

A spectrum of a whole sentence mixes every vowel into one smear. You cut the wave into short frames so the tract stays nearly fixed inside each one. A common choice is a window of 25 milliseconds and a hop of 10 milliseconds. At 16 kilohertz the window holds 400 samples and the hop holds 160. You multiply the 400

samples by a Hamming window so the frame edges fade, which keeps the abrupt cut from inventing fake high frequencies.

Count the frames before you train anything. One second at 16 kilohertz is 16000 samples. The first frame uses samples 0 through 399. Each hop moves 160 samples. The last legal start is 16000 minus 400, which is 15600, and 15600 divided by 160 is 97.5. You can start a frame only at whole hops, so there are 97 hops after the first frame, and 98 frames in all. A file one sample shorter can drop that last frame.

How one second becomes 98 frames.

1. Take a 400-sample window

2. Hop forward 160 samples

3. Stop at start index 15600

4. Count 98 legal frames

Frame count for one second

1. The sample rate is 16 kilohertz, so 1 second is 16000 samples. The window is 400 samples and the hop is 160.

2. The last start index is 16000 minus 400, so 15600. Dividing 15600 by 160 gives 97.5, and a start index must be a whole number of hops.

3. You get the first frame plus 97 later starts, which is 98 frames. A file of 15999 samples cannot host that last start, so it can have 97 frames instead.

Problem 1. At 16 kilohertz, with a 400-sample window and a 160-sample hop, how many frames are in 1 second?

Answer. 98. The last legal start is 15600, 15600 divided by 160 is 97.5, and only 97 whole hops fit after the first frame.

Problem 2. Why multiply the 400 samples by a Hamming window instead of cutting the wave with a hard edge?

Answer. A hard cut invents high frequencies that were not in the tract. The Hamming window fades the frame edges so the spectrum you measure is closer to the short-time vowel.

Lecture 4 · 90 minutes

A mel bank and a cepstrum

Inside one frame you build a mel spectrum, then compress it. Take the power in each of 26 triangles, add a floor of 1e-10 so you never take a log of 0, and take the log. The log turns a multiplicative tilt of the spectrum into a shift you can subtract later. You then apply a discrete cosine transform and keep the first 13 coefficients. Those 13 numbers are the cepstral vector for that frame.

You append how the vector is moving. Subtract the previous frame's 13 coefficients from the next frame's and divide by 2 hops, which is a delta. Do that again on the deltas to get a delta-delta. Stacking static, delta, and delta-delta yields 39 numbers per frame. A static vector cannot tell a rising formant from a falling one. The deltas

can, at the price of 2 extra frames of context at each edge, where you may repeat the end frame.

From filter energies to 39 numbers.

26 log filter energies

13 cosine coefficients

13 deltas

13 delta-deltas

A two-sided difference

1. One cepstral coordinate is 1.0 on the previous frame and 1.4 on the next frame. The central frame is the one you are labeling.

2. The delta is (1.4 minus 1.0) divided by 2, so 0.2. You do not use the central value in this particular difference.

3. Stacking 13 static values, 13 deltas, and 13 delta-deltas yields 39 numbers. At the first frame you have no previous frame, so you repeat the end value rather than inventing a slope of 0 without saying so.

Problem 1. Previous value 1.0, next value 1.4. What is the delta if you divide the difference by 2?

Answer. 0.2. The difference is 0.4, and 0.4 divided by 2 is 0.2.

Problem 2. Why keep 39 numbers instead of the 13 static cepstra alone?

Answer. The static 13 cannot tell a rising formant from a falling one. Deltas and delta-deltas add that motion, at the cost of 2 frames of context at each edge.

Lecture 5 · 90 minutes

A hidden chain for a phone

A phone is not one frame. You model it as a small chain of hidden states, often 3 states for a phone, and each state emits a feature vector with some density. Transitions between states carry probabilities that sum to 1. A left-to-right chain lets you stay in a state, which stretches the phone in time, or move forward. You do not observe the state. You observe the 39-dimensional frame and must infer the path.

Viterbi keeps the best path into each state at each frame. You multiply incoming score, transition, and emission, or add logs so the product does not hit 0. At the last of 98 frames you take the max score and walk the stored backpointers. A path that stays in one of 3 states for every frame is legal in a left-to-right chain only if self-loops exist. That walk is the recognized state sequence.

Viterbi keeps one best path per state.

1. Multiply transition and emission

2. Keep the best incoming path

3. Store a backpointer

4. Walk back from the last frame

Two states at frame 2

1. Start probabilities are 0.5 and 0.5. Frame-1 emissions are 0.5 for A and 0.2 for B, so the scores are 0.25 and 0.10.

2. Transitions: A to A is 0.6, A to B is 0.4, B to A is 0.4, B to B is 0.6. Frame-2 emissions are 0.4 for A and 0.5 for B.

3. Into A, the path from A scores 0.25 times 0.6 times 0.4, so 0.06, and the path from B scores 0.10 times 0.4 times 0.4, so 0.016. Into B, the path from A scores 0.05 and the path from B scores 0.03. You pick A with score 0.06. The path is A then A.

Problem 1. Using the worked transitions and emissions, what is the best score at frame 2, and which state has it?

Answer. State A with score 0.06. The competing path into B scores 0.05. The backpointer for A is A, so the path is A then A.

Problem 2. A phone has 3 states and no self-loop. What is the longest frame sequence that phone can cover?

Answer. 3 frames. Without a self-loop a left-to-right chain cannot stay, so it cannot stretch past one frame per state.

Lecture 6 · 90 minutes

A network that scores frames

A neural acoustic model replaces the emission table with a network. You feed 40 log mel bins, plus 2 frames on each side. For a single frame that is 5 times 40, or 200 inputs. The network ends in a softmax over the phone set. If you have 40 phones, the output is 40 probabilities that sum to 1. You train it to put mass on the labeled phone for that frame.

Cross entropy falls when the probability on the correct phone rises. Between two models on the same frame, you prefer the one that assigns 3/4 to the correct phone over the one that assigns 1/2. The hidden chain can stay: the network supplies the emission, and Viterbi still multiplies by transitions. A 3-state phone needs 3 outputs, or you tie them and share 1. A language model can still veto a word the acoustics liked.

Frames in, phone probabilities out.

Stack 5 frames of 40 -> Run the network -> Softmax over 40 phones -> Hand scores to Viterbi

Two models, one correct phone

1. Model 1 puts probability 1/2 on the labeled phone. Model 2 puts 3/4 on that same phone. The other probability mass is spread over the remaining phones and sums with these to 1.

2. Cross entropy uses the negative log of the correct probability. Because 3/4 is larger than 1/2, model 2 has the smaller loss on this frame.

3. You still multiply the chosen probability by a transition inside Viterbi. An acoustic 3/4 on the wrong

word can lose to a word sequence the language model scores higher.

Problem 1. One model assigns 1/2 to the correct phone and another assigns 3/4. Which has the lower cross entropy on that frame?

Answer. The model that assigns 3/4. Cross entropy falls as the probability on the correct phone rises.

Problem 2. The acoustic model likes a word and the language model calls that word impossible. Which score is allowed to veto?

Answer. The language model can veto the word. Acoustic probability alone is not the recognizer. Viterbi still multiplies emissions by transitions, and the language model scores the word sequence.

Lecture 7 · 90 minutes

Change duration without changing the vowel

Synthesis runs the source-filter model forward. You place 100 impulses in one second and pass them through the vowel filters. For 125 hertz you place 125 impulses and leave those filters fixed. The vowel should not move. To stretch time, repeat a frame for 2 hops instead of 1 so duration doubles while pitch stays near 100 hertz. Pitch and vowel are separate controls, and you should not move both when you meant one.

Modification of a recorded utterance uses the same split. You mark pitch periods, about 160 samples apart at 16 kilohertz if the pitch is 100 hertz, because 16000 divided by 100 is 160. Overlap-add can then repeat or drop periods. Dropping every other period raises the perceived rate, but if you also resample the wave by 2 you changed duration and pitch together. Say which of the two you intended before you touch the samples.

Two edits that are not the same.

Pitch only

100 impulses become 125

Filters stay fixed

Vowel should not move

Duration only

Use a frame for 2 hops

Impulse rate stays

Local time doubles

One frame, two hops

1. The hop is 10 milliseconds and the pitch is 100 hertz, so a one-second vowel would hold 100 impulses and about 98 analysis frames.

2. You repeat one vowel frame so it is used for 2 hops, 20 milliseconds, instead of 1 hop. Local duration doubles.

3. The impulse spacing inside the frame stays at 160 samples, which is 100 hertz at 16 kilohertz. The

vowel filters are untouched. You changed time, not pitch.

Problem 1. At 16 kilohertz and 100 hertz pitch, how many samples apart are the pitch marks?

Answer. 160. You divide 16000 samples per second by 100 impulses per second.

Problem 2. You resample the whole wave by 2 and also drop every other pitch period. Why is that a confused edit?

Answer. Resampling by 2 changes duration and pitch together. Dropping periods is a separate rate change. You have moved both knobs, so you cannot tell which one produced the vowel you hear.

Lecture 8 · 90 minutes

A front end you can trace

You can now build a front end and say what each stage did. From one second at 16 kilohertz you cut about 98 frames, map each through 26 mel filters to 13 cepstra, and stack deltas to 39. You can score a phone chain with Viterbi or replace the emission with a softmax over 40 phones. You can place 100 impulses through a fixed filter to hold a vowel, or repeat a frame across 2 hops to stretch time.

You still owe a count, not a spectrogram you liked. On 20 held-out seconds, count frames where the argmax phone is wrong, and count words the best Viterbi path gets wrong. If frames are wrong but words are right, the language model is hiding a weak acoustic score. If both are high, look at the 98-frame count and the 39-vector before you add layers. You should be able to change pitch by moving only the impulse rate.

The path from a wave to a score.

98 frames from one second

13 cepstra, or 39 with motion

Viterbi or a 40-way softmax

A count of frame errors

How many numbers one second holds

1. One second at the settings above is 98 frames. Each frame, with deltas, is 39 coefficients.

2. The matrix holds 98 times 39 numbers. That product is 3822.

3. If you had kept only the 13 static cepstra, the matrix would be 98 times 13, which is 1274. The extra 2548 numbers are the deltas and delta-deltas, 2 times 1274. You should know which matrix you handed to the recognizer.

Problem 1. On 20 held-out seconds the frame error is high and the word error is low. What is covering the acoustic mistakes?

Answer. The language model. It can pick a likely word even when many frame posteriors are wrong. Look at the acoustic scores before you add more layers.

Problem 2. Combine framing and cepstra. One second is 98 frames of 39 coefficients. How many numbers is that, and why is it not 98 times 13?

Answer. 3822. The 13 static cepstra are stacked with 13 deltas and 13 delta-deltas, so each frame is 39 numbers, and 98 times 39 is 3822. The static-only matrix would be 1274.